

Skin It To Win It

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Abstract

Skin It to Win It is a prototype for a skincare product recommendation and comparison app with social functionality. The primary goal was to make skincare research less burdensome for users and to help them create a routine that meets their needs using what is locally available. Given the available time, it was not feasible to produce a complete application, but an earnest effort was made to demonstrate intended functionality. Ultimately, the deliverables include a SQL database designed with the app's functionality in mind and populated with real Sephora product data, a high-fidelity Figma prototype to demonstrate the intended look and feel, and a web app that demonstrates the initial user flow and delivers recommendations from the database based on priorities extracted during the onboarding process.

Introduction

Skincare is a common part of daily life, but selecting the right products is still a challenge for many. There are numerous cleansers, serums, toners, moisturizers, and treatment creams available that can look very similar while being quite different in terms of their formulas and purposes. Despite the availability of these products, it might not be so easy to understand what they are meant to do. Most products come with information that focuses primarily on brand positioning and marketing, rather than explaining how to use the product effectively.

The issue aligns with the existing literature on the topic. The findings from previous studies reveal that cosmetics and skincare products are influenced by branding, ingredients, and skin conditions, whereas people lack full awareness of a product's effectiveness and potential dangers (Krishnan et al. 48–49). Also, research on consumer choices in skincare reveals that customers make purchasing decisions based on several factors, including brand, durability, efficacy, cost, and scent (Kamwendo and Maharaj 82–83). Meanwhile, it is worth noting that the functional properties of cosmetic ingredients are significant, as these substances directly affect the performance of a skincare product (Sharma et al. 1–3).

The survey conducted as part of the project itself highlighted this problem. The month-long online survey, with 170 respondents, showed that some users did not understand ingredients or labels and did not feel confident even after researching.

Skin It to Win It was developed to address this need. The three main parts of Skin It to Win It are: a structured skincare products database; a skin quiz that helps identify the user's needs and preferences; and a user interface that enables searching for products and comparing them. What distinguishes the project from most skincare apps currently available is its community component, which will allow people to share their experiences beyond product data. The objective wasn't creating a commercial app in a semester's timeframe, but rather building a solid project through research, design, prototyping, and programming that would be useful to the everyday user.

Positionality

The approach adopted by the project's developers significantly influenced its nature. Firstly, the team did not approach the issue from a medical angle but rather as students who wanted to improve the accessibility of information on skincare products and methods. This design perspective was also influenced by reviewing existing apps such as Lumi Skin, whose clean, minimal user interface (UI) and user experience (UX) helped clarify the value of simplicity, readability, and immediate usability in skincare design. Secondly, because the survey showed that people often use their peers' experiences to help them decide which skincare product to buy or try, it became crucial to make interaction within the community the core idea, rather than an option.

However, one needs to recognize the constraints of this approach as well. It is neither intended to diagnose skin conditions nor to replace dermatologists' expertise. People vary in terms of skin type, sensitivities, lifestyles, medical histories, and objectives, and a student-developed prototype will have limitations in addressing all these variables. Although the study of consumer behavior helped recognize common areas of misunderstanding and desire, it is unrealistic to claim that this model reflects the experiences of all skincare users.

This study was not limited to the mere creation of a technology-oriented solution; it integrated the survey and prototype design into a single research effort.

Background

The literature highlights the impact of consumer needs, attributes, ingredients, and confidence in making skincare decisions. Studies on cosmetic use have indicated that consumers tend to be swayed by branding, ingredients, and skin issues when buying cosmetics, unaware of their utility and potential risks (Krishnan et al. 48–49). In a study on skincare preferences, Kamwendo and Maharaj have indicated that consumers tend to consider multiple attributes when evaluating skincare products, particularly durability, brand, effectiveness, pricing, and fragrance (82–83).

Understanding cosmetic ingredients is therefore essential. As Sharma et al. argue, cosmetics contain functional ingredients such as emulsifiers, preservatives, humectants, emollients, antioxidants, and UV filters. These affect the performance of the cosmetic products and their outcome (1–3). Besides, according to Sharma et al., consumer awareness and a desire for safer, non-irritating cosmetics have highlighted the importance of ingredient knowledge (1–2).

Recommendation research adds another relevant dimension. According to Chen, skincare recommendations are challenging, as false recommendations may lead to irritation, allergies, or even worsen the skin's condition (456–57). Additionally, according to Chen, successful skincare recommendations require multidimensional information about both user and product attributes, such as skin type, sensitivity, concerns, ingredients, active substances, and irritants (457). Thus, the proposed system, with its skin quiz and structured product characteristics, appears justified.

Community structure and user profile design are also applicable topics. As shown by Hartzler et al., patients in online health communities often seek recommendations from other community members who have had the same experience. However, this process might become challenging when dealing with poorly filled user profiles. The research shows that relevant user profiles, summaries, and cues help better

identify relevance (Hartzler et al. 15:1–15:4). Although Hartzler et al. explore health forums and do not discuss skincare communities specifically, the findings remain useful for this proposed application.

Finally, Xu et al. explain that, through structured feedback during iterative design, deep and surface revisions can be made, making revision an essential element of the design process (1637–38). This supports the development method used in Skin It to Win It, in which the low-fidelity drawings were continuously revised until a high-fidelity prototype was developed.

The analysis of applications on the App Store, such as Lumi Skin, OnSkin, INCI Beauty, Skin Care Safe Scanner, Skincare Scanner, and Yuka, revealed that although there were many ingredient-checking applications, most lacked strong user interaction and community involvement. In addition, while services like Curology can offer more tailored recommendations through expert consultations, they do not necessarily provide the instant, easy guidance that some consumers may need when no medical knowledge is required. This contrast is illustrated in Figure 1, which shows Curology’s emphasis on personalized skincare through a more professional and subscription-based model. While this model offers value through personalization, it also helped inspire the idea of creating a more accessible and lower-cost alternative for users who want guidance without necessarily needing a paid expert service.

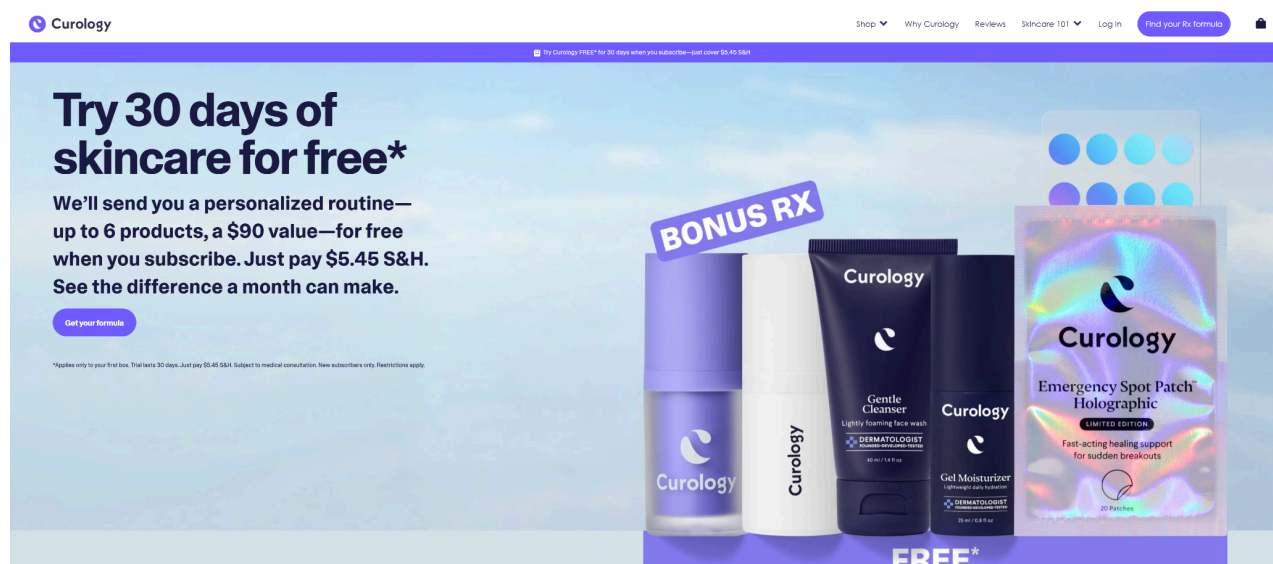


Figure 1. Curology interface showing a personalized skincare service model with a more professional and subscription-based structure.

Objectives

A primary objective was to determine whether, in aggregate, Skin It to Win It was even necessary or desirable to a potential consumer base. But user research was necessary to ensure this was not a minority opinion. Ascertaining the project's viability and demand, as well as the functionality of individual components, would inform development.

The design objective was to create and demonstrate the intended look and feel. Through iterative prototyping, the project would develop a comprehensive slice of the app that could be interacted with to explore the actions the average user could perform and the benefits available to them.

The database engineer's objective was to transform existing product data into a database that accurately encoded the links between products, their ingredients, the purposes of those ingredients, and the goals users might have for their skincare routine. For the app to truly be data-driven, those relationships had to be enumerated and easily traceable with simple queries. The database should also be efficient and fast to minimize user frustration and decrease data integration overhead.

Project requirements also dictated a demonstration of the intended active functionality beyond facsimile prototypes, so that individuals unfamiliar with the database could envision its inner workings and how products would be recommended in the finished application. The "skin quiz" feature, which would automatically assign each user priority weights that would, in turn, be used to recommend products, was a required component of this demonstration. As such, this component could be considered another prototype, but on a functional level rather than a visual level.

Methodology

User Research

This project was developed through multiple stages of research, UI/UX design, database building, and functional prototyping. Initially, user needs were identified through research. Using Qualtrics, a one-month survey was conducted with 170 participants, who shared their insights on skincare knowledge

and decision-making, as well as the app features they would like to see. These insights were then turned into an infographic that outlines pain points, user needs, and the project's proposed solution.

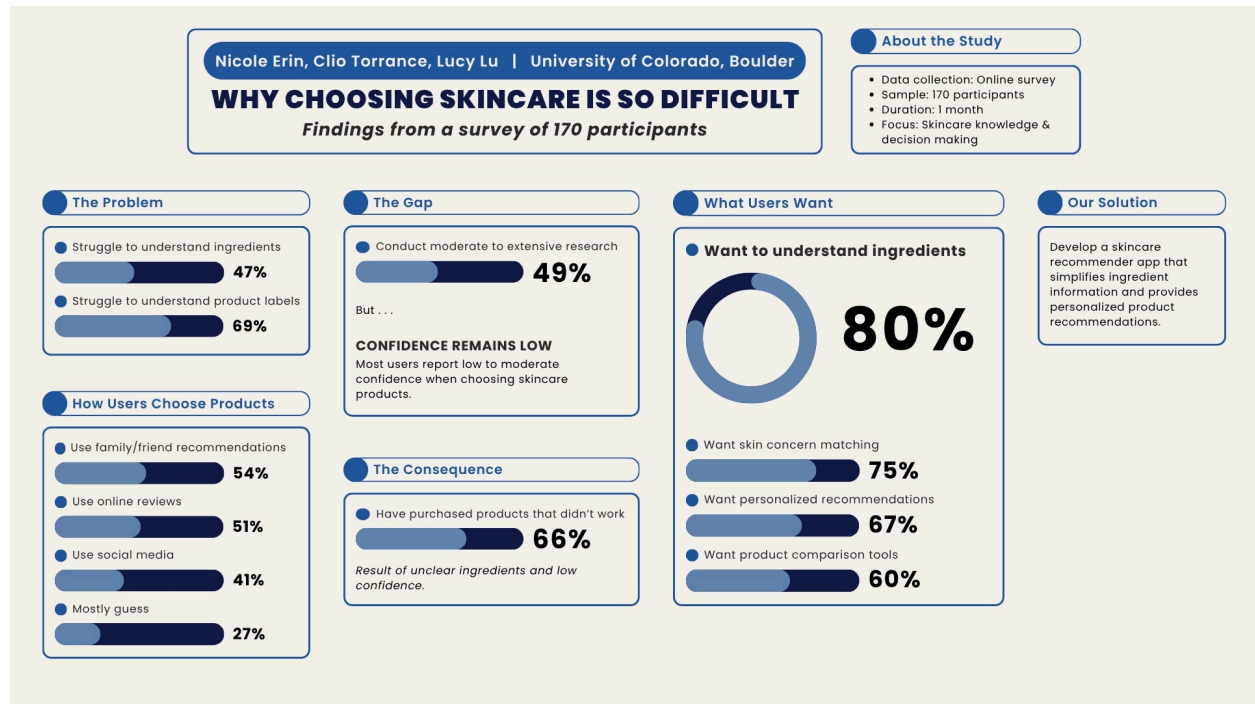
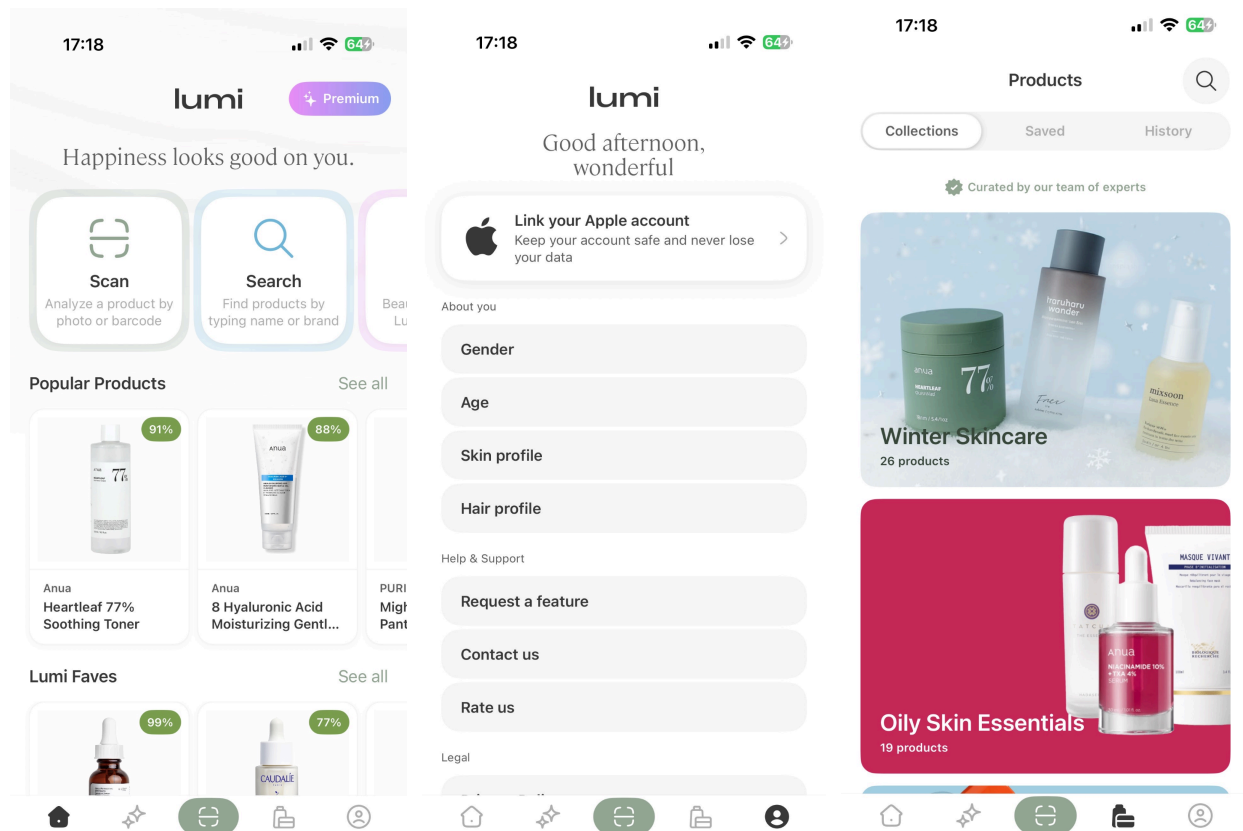


Figure 2. Infographic summarizing project justification survey results

Additionally, the existing skincare and ingredient-analysis applications were examined on the App Store. Lumi Skin, OnSkin, INCI Beauty, Skin Care Safe Scanner, Skincare Scanner, and Yuka were compared to identify common features across existing applications.



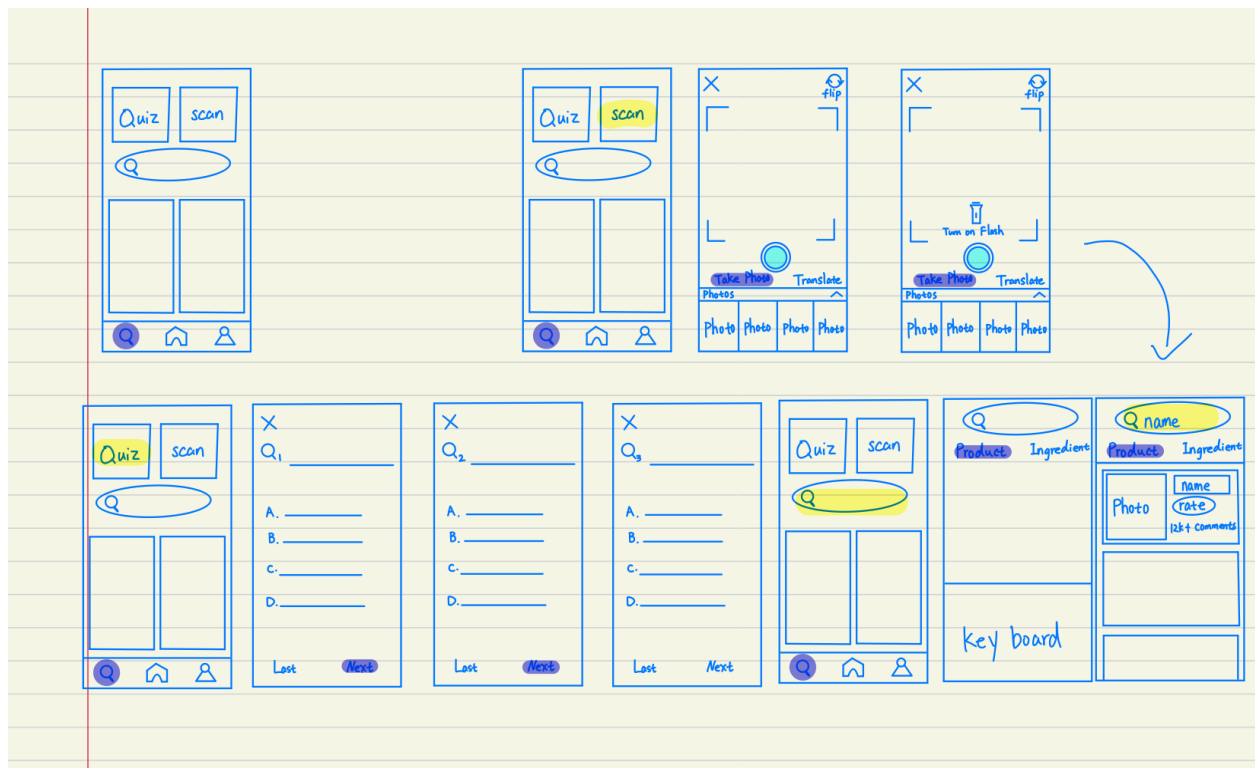
Figures 3–5. Lumi Skin interface examples featuring a clean, minimal UI/UX design, immediate access to major functions such as scan and search, and a simplified personalization layout.

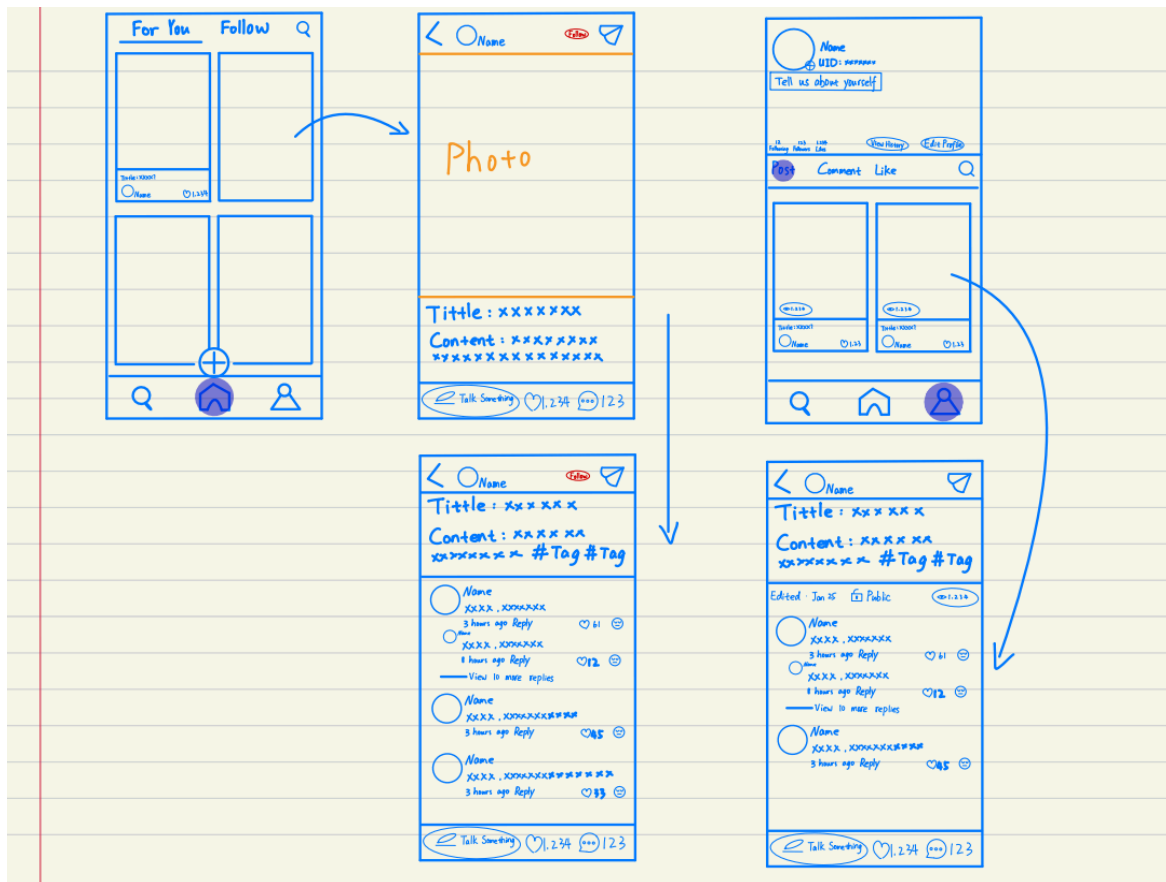
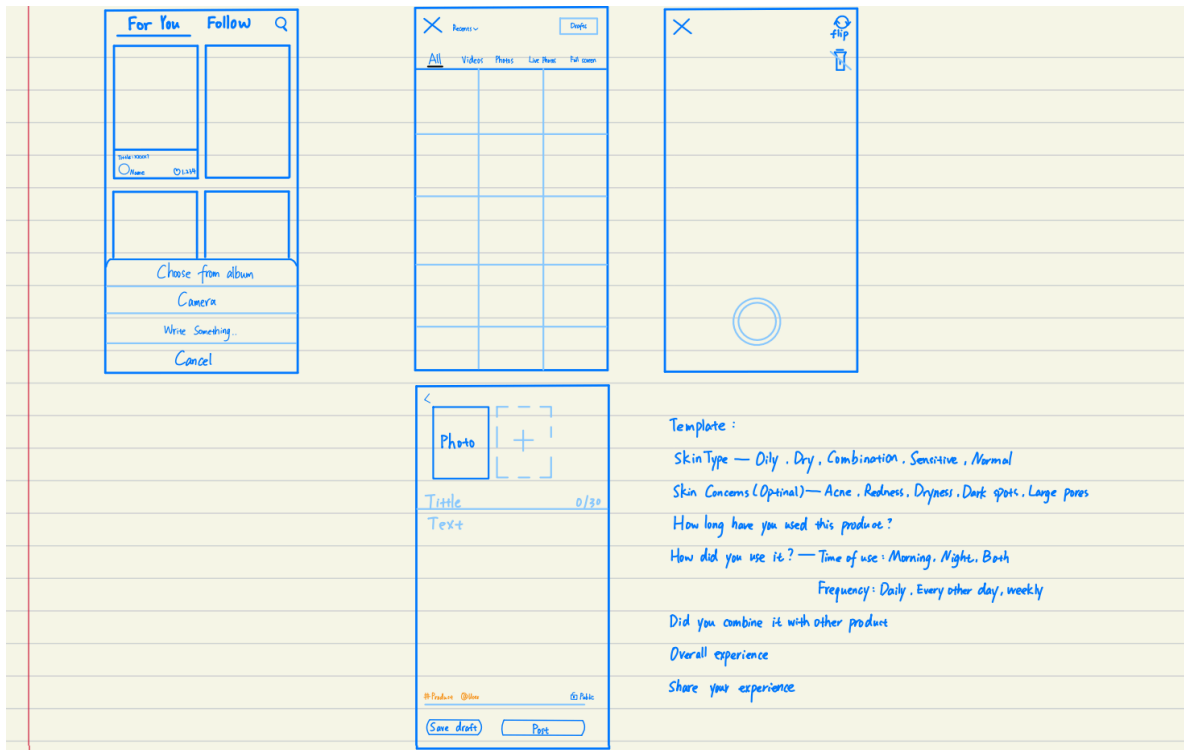
Low-Fidelity Prototyping

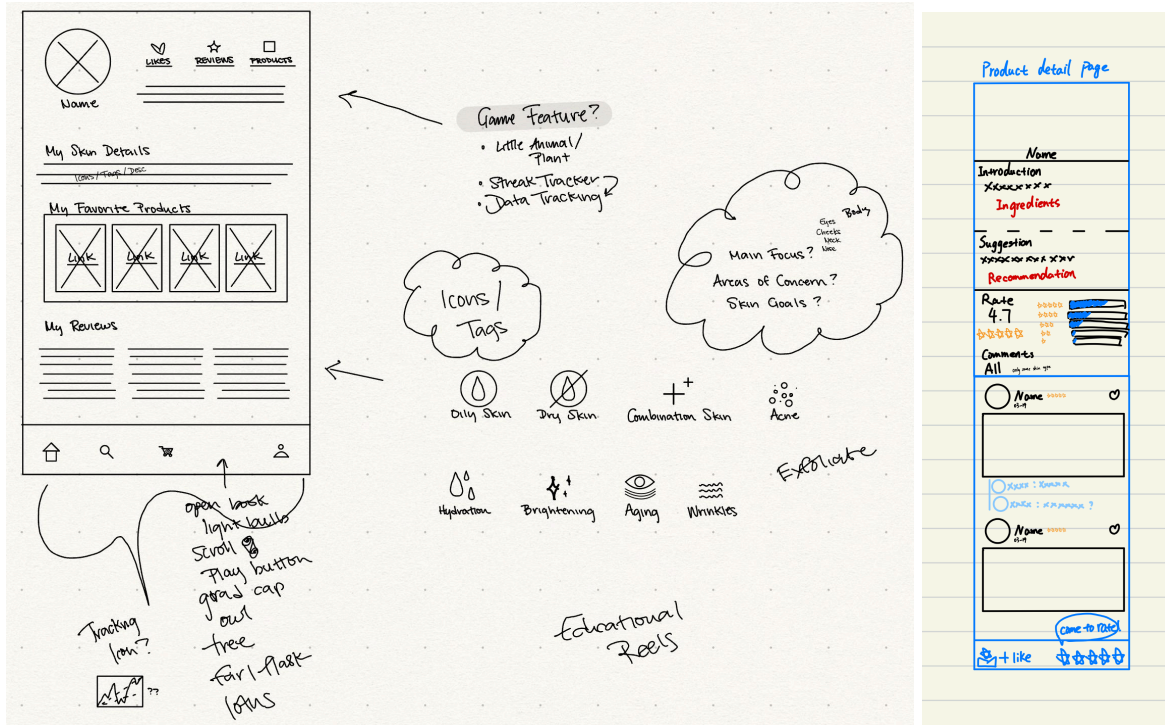
The low-fidelity prototyping phase focused on defining the app's primary structure and identifying the most critical entry points for app users. Sketches created during this phase were strongly influenced by Lumi Skin's website homepage design, which effectively presented key features as soon as the user opened the application. It made it clear that quizzes, scans, and searches are crucial for creating a convenient and efficient user experience from the very first moment.

The low-fidelity phase also progressed from the homepage to other parts of the interface. These wireframes considered user interaction with other sections within shared posts, personal tracking, and product information.

This is the point at which some key design questions came to light. For example, it was clear at this point that the search flow was somewhat detached from both the home page and the user's profile page. It also became necessary to determine which ingredient details should go straight to the product detail page and which should be delegated to a new, more dedicated ingredient detail page. The figures below (Figures 5–9) demonstrate some of the work done during this lower-fidelity stage, when the main aim was to explore functionality, navigation, and layout rather than visual design. This iterative process aligns with Xu et al.'s argument that structured feedback can lead to both deep and surface-level design revisions over time (1637–38). Quiz question creation and score evaluation were also completed in the low-fidelity phase as part of the application's personalization process.







Figures 6–10. Low-fidelity prototype wireframes showing the early exploration of app structure, user flow, and feature relationships among quiz, search, scan, profile, product detail, and community functions.

High-Fidelity Prototyping

The high-fidelity prototyping phase refined the design created during the previous stage, making it even more refined, consistent, and user-oriented.

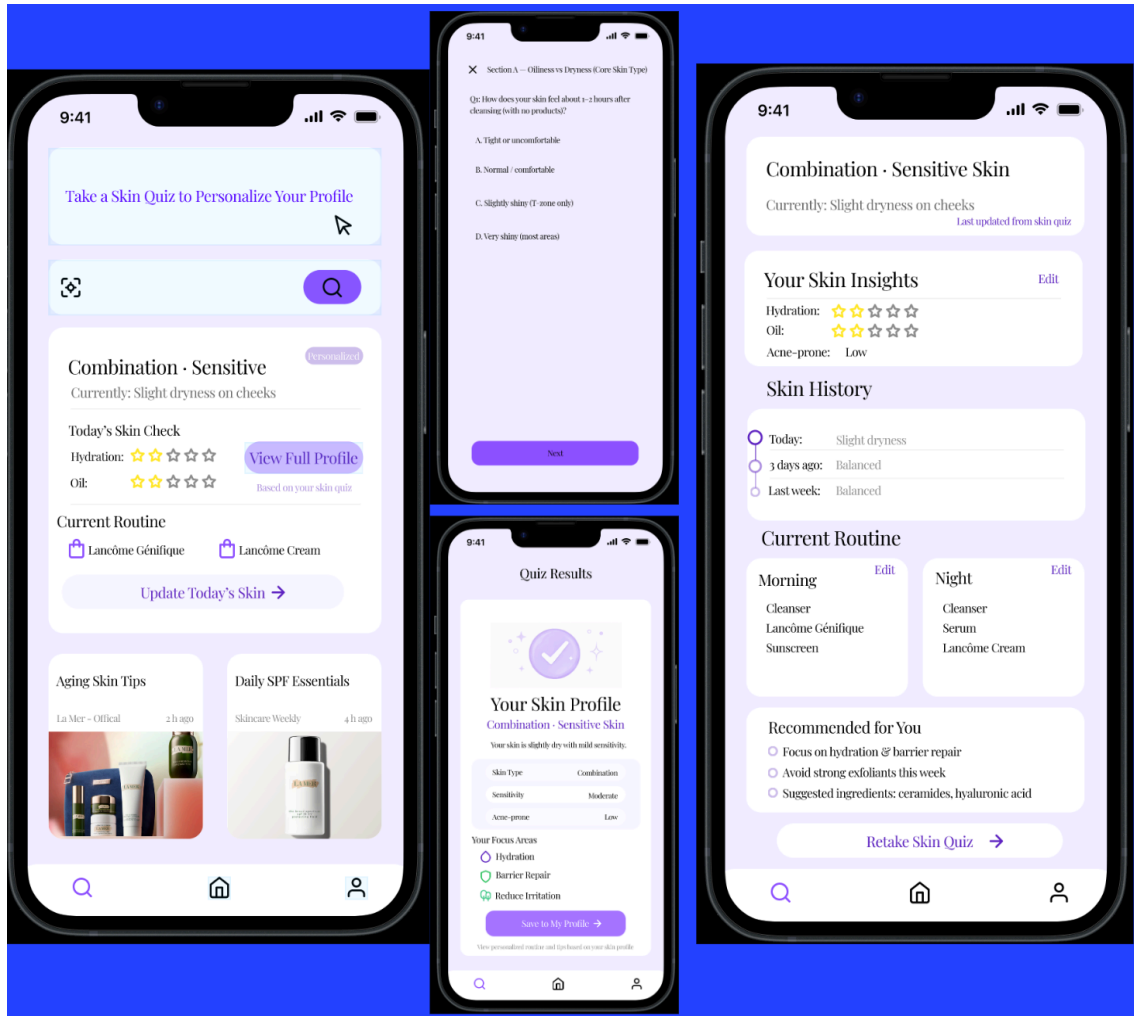
A major change was the redesign of the search page. The scanning function was removed from the homepage and embedded in the left side of the search bar. At the same time, a quick link to the quiz highlighting the concept of personalization was kept and made easier for users to find. A short form of a personalized skin profile, with current information on one's skin condition and the products used, was also embedded on the homepage, eliminating the need for an extra click to access the profile.

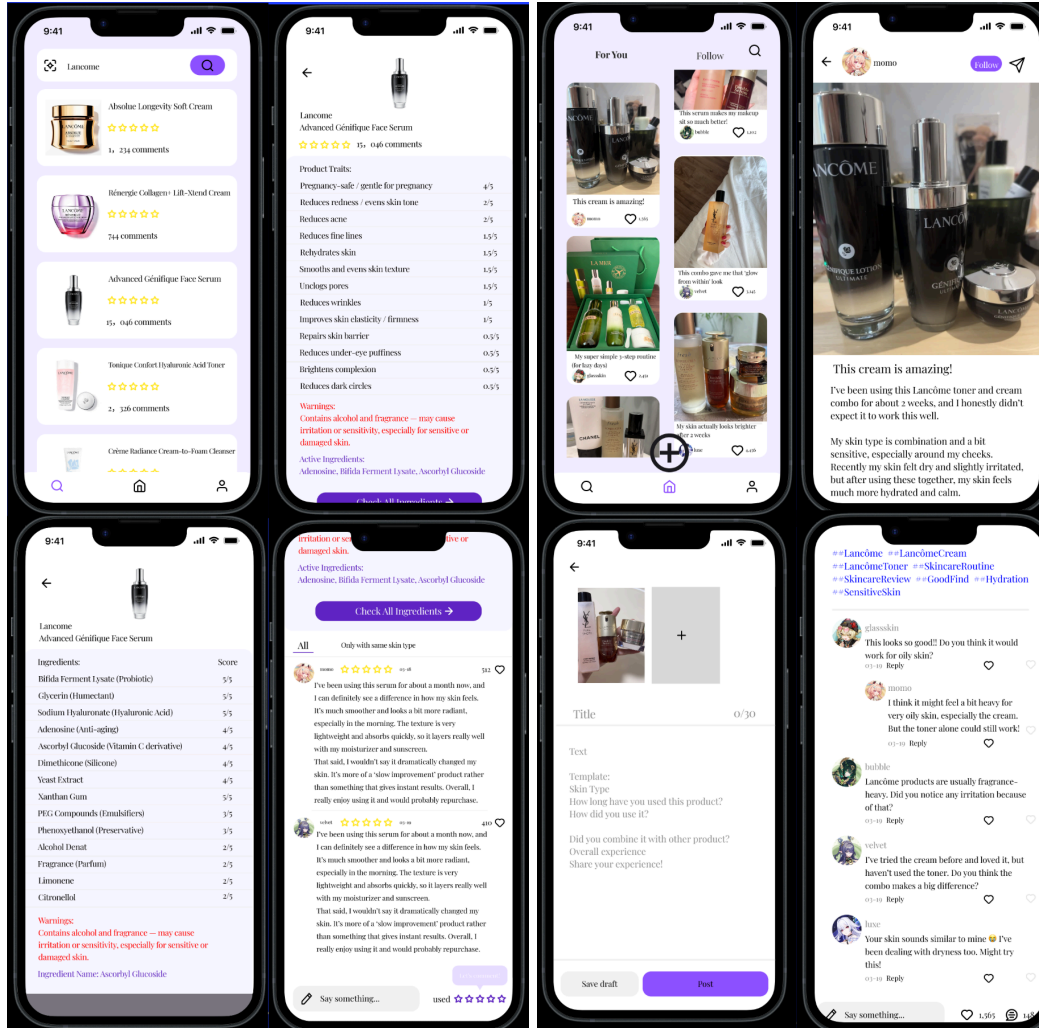
The user profile was further developed to become an independent, detailed page. In the high-fidelity prototype, the page would include information such as the user's skin type, the day's skin history, the user's current skincare routine, and personal recommendations/reminders. Moreover, more

connections were established between the quiz and the profile, ensuring that users' quiz-derived skin profiles matched their saved profile, enabling them to customize their skincare routine.

Another important improvement made as a result of this prototype's high-fidelity was in the product search page and product details page. The high-fidelity design of the search results interface allows users to search for either a brand or a product and then switch to the product details page after finding what they are interested in. Unlike the low-fidelity design, the high-fidelity product detail page simplified the rating section, allowing more space for the product's functionality, potential risks, and reviews from other users with the same skin type. This page was designed with the available ingredient literature in mind. Some users prefer to read a lot more about the ingredients than others do (Sharma et al. 1–3).

However, the community aspect became one of the most crucial elements within the design of the high-fidelity prototype. The initial ideas proposed during the creation of the low-fidelity prototype for posts, comments, likes, and user following have been used and improved for consistency and structure. For example, a guided structure has been added to help new users understand the kind of information they should include when making a post, including the type of skin, its issues, time and frequency of use, and overall impression. It is worth noting that the emphasis on community in this case is consistent with the research by Hartzler et al., who found that profiles, summaries, and contextual clues help individuals find relevant experiences in online health communities (15:1–15:4).





Figures 11–13. High-Fidelity Prototype Screens depicting a more refined interface that includes improvements to the homepage, linking quiz and profiles, product detail optimization, ingredient warnings, and community interactions.

Backend Development

PostgreSQL was selected as the database component for Skin It to Win It after some research. It was deemed most suitable for three reasons: 1) seeming comparative ease of setup on Linux systems, which the database engineer preferred; 2) certain extensions to the SQL specification implemented by PostgreSQL were desirable for more efficient query construction; 3) ready compatibility with the Python programming language through the simple yet robust psycopg2 module.

The primary goal of this database was to encode relationships among products, ingredients, traits, and the purposes for which those traits would be useful. With these links represented, finding a way from the user's skincare goals to products useful in those contexts would be as simple as performing a series of filtering joins, allowing recommender logic to be baked into the data itself.

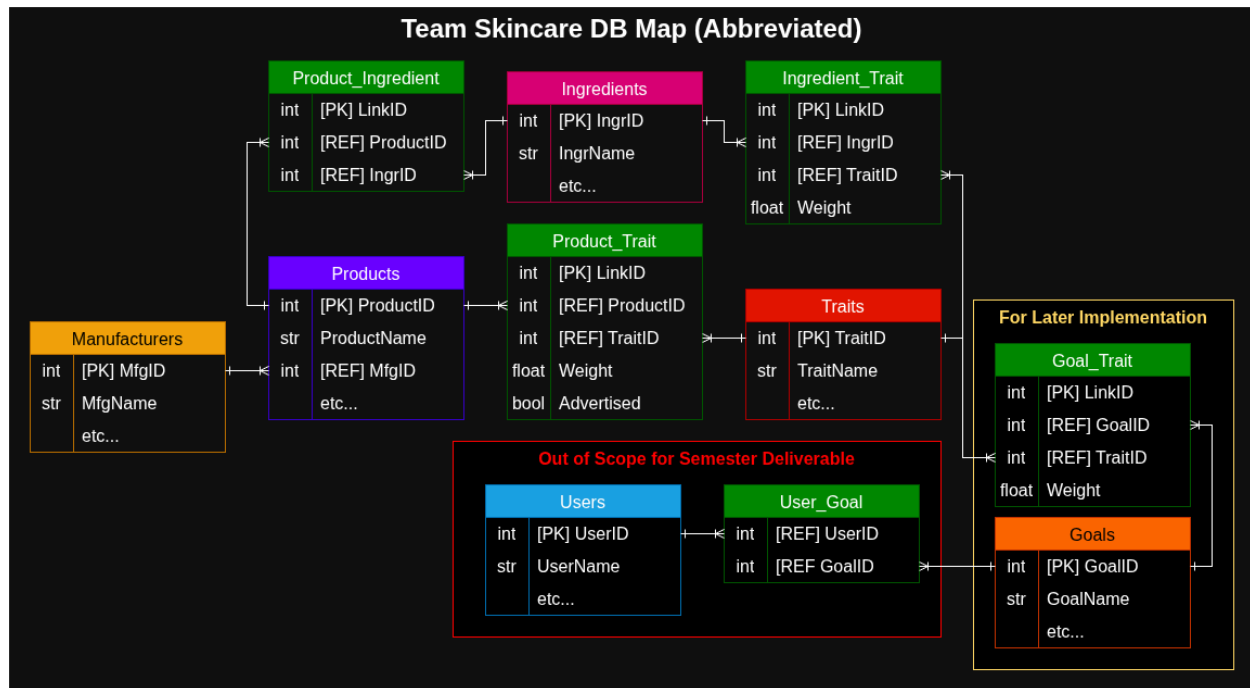


Figure 14. First draft of database structure.

The initial plan for the database structure (fig. 13) focused on table relationships and did not attempt to enumerate every required field. Discussion was still ongoing at this point about which functionality would be most essential for the prototypes and deliverables, given the available timeframe. Cross-ref tables were given their own primary key column, as the database engineer envisioned joins combining the trait-ingredient-product link and the trait-product link.

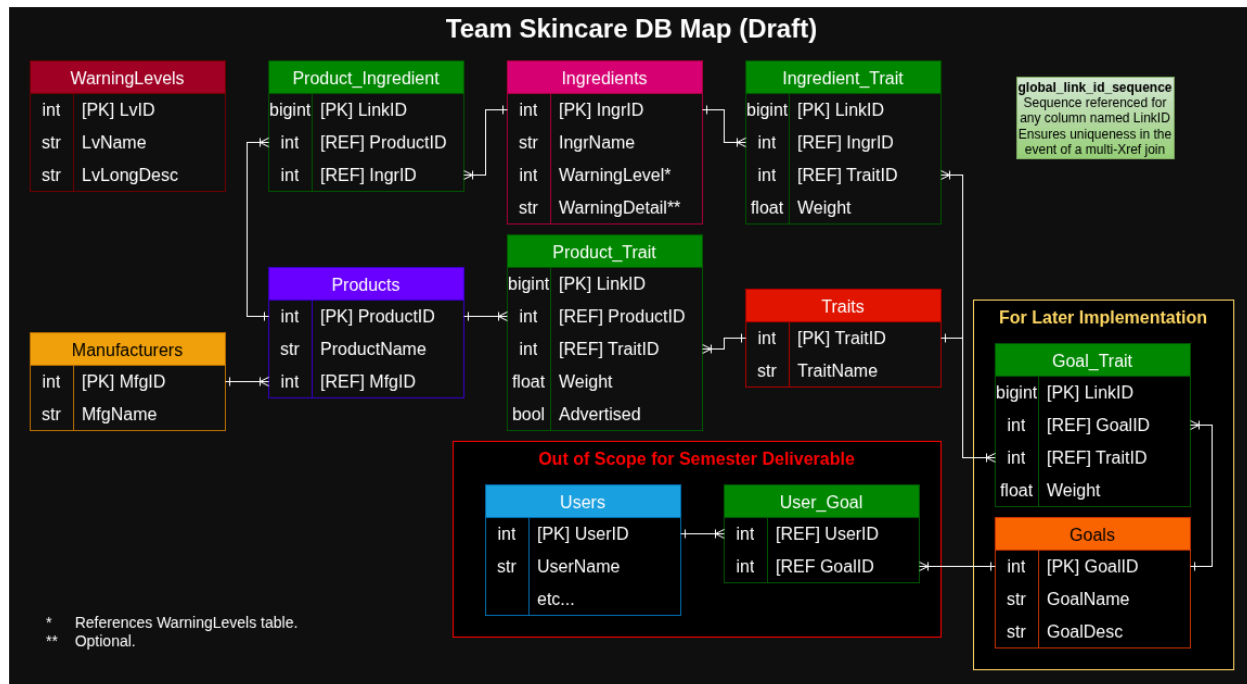


Figure 15. Second draft of database structure.

Once PostgreSQL was up and running, the database structure was expanded to include ingredient warning levels (fig. 14), which were added to the specification for what information was required around this time.

Database population then commenced, using the Python module Pandas to rearrange tables of real product and ingredient information into the database's format, which was then loaded into PostgreSQL via Python queries.

Recurring user configuration issues impeded this process and prevented external access to the database, impairing front-end development. The decision was eventually made to swap to Supabase, a PostgreSQL hosting service with a limited free plan that would be sufficient for this project's scope.

After the swap, adding data to the database became a more streamlined process, and with this newfound time, revisions were made to the database structure to include more information on warning levels and eliminate the unnecessary global cross-ref key sequence (fig. 15).

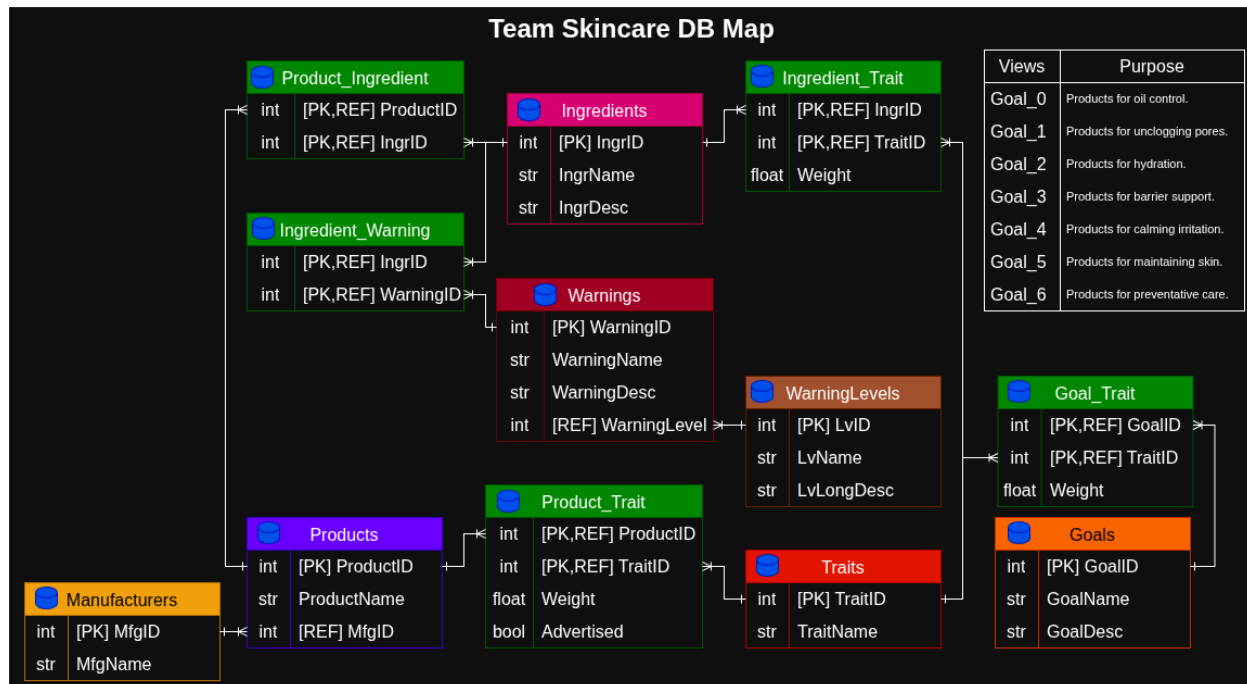


Figure 16. Final database structure.

Finally, to aid front-end development, views were created that performed the necessary joins to locate products appropriate for seven specific skin goals. The sum of the Weight fields in the various links to each product was used to roughly evaluate comparative product suitability.

Web App Development

The development of the web application component was not originally part of this project due to time constraints. However, after much deliberation among supervisors and mentors, it became clear that the project needed an interactive layer that translated research and database logic into a useful, functional user-facing tool. The primary goal when developing this web app was to deliver a polished, reliable experience that could be easily demonstrated. To achieve this, the frontend interface, quiz and scoring logic, product recommendation system, and deployment infrastructure were developed in parallel.

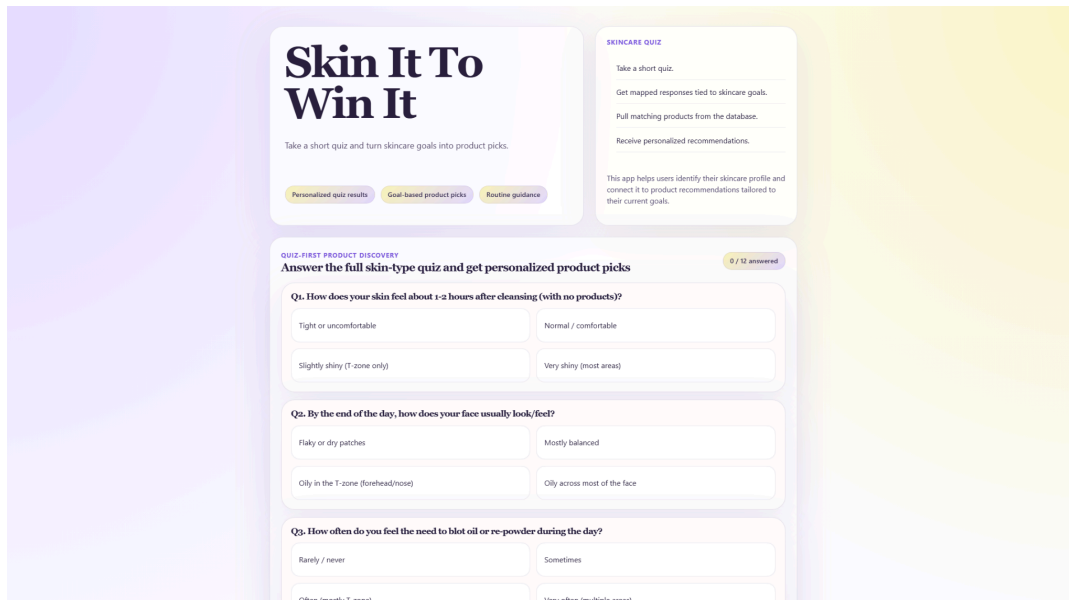


Figure 17. Homepage of the Skin It to Win It web application, demonstrating the overall interface and visual design system.

From the beginning, the app was designed to be visually consistent and inviting, with a welcoming color palette and clear homepage structure. The quiz was implemented with a straightforward, step-by-step flow, removing unnecessary technical details and internal labels to prioritize user clarity.

QUIZ-FIRST PRODUCT DISCOVERY
Answer the full skin-type quiz and get personalized product picks 12 / 12 answered

Q1. How does your skin feel about 1-2 hours after cleansing (with no products)?

☐ Tight or uncomfortable	☐ Normal / comfortable
☐ Slightly shiny (T-zone only)	☐ Very shiny (most areas)

Q2. By the end of the day, how does your face usually look/feel?

☐ Flaky or dry patches	☐ Mostly balanced
☐ Oily in the T-zone (forehead/nose)	☐ Oily across most of the face

Q3. How often do you feel the need to blot oil or re-powder during the day?

☐ Rarely / never	☐ Sometimes
☐ Often (mostly T-zone)	☐ Very often (multiple areas)

Q4. When you apply moisturizer, what happens most often?

☐ My skin absorbs it quickly and still feels dry	☐ It feels comfortable and balanced
☐ It can feel heavy or greasy sometimes	☐ It almost always feels too greasy

Q5. How often does your skin sting, burn, or itch after skincare products?

☐ Never	☐ Rarely
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Q9. Do you get clogged pores or blackheads easily?

☐ Rarely	☐ Sometimes
☐ Often	☐ Very often

Q10. What is your current climate most of the year?

☐ Dry / cold	☐ Mild / balanced
☐ Humid / hot	☐ Changes a lot (seasonal extremes)

Q11. How many skincare steps do you usually use (AM or PM)?

☐ 0-1 (very simple)	☐ 2-3
☐ 4-5	☐ 6+ (many active products)

Q12. What is your main skincare goal right now?

☐ Hydration / repair barrier	☐ Oil control
☐ Clear acne / reduce breakouts	☐ Calm redness / reduce sensitivity
☐ Brightening / even tone	☐ Anti-aging / smoothing texture

[See Recommendations](#) [Clear Answers](#)

Figures 18-19. Step-by-step skin quiz interface designed to guide users through personalized onboarding.

When users submit their answers, the app calculates their skin profile using a scoring model based on key dimensions, including oiliness, dryness, sensitivity, and acne-proneness. These results then drive a set of recommendation goals, which are used behind the scenes to match users with relevant skincare products and routines.

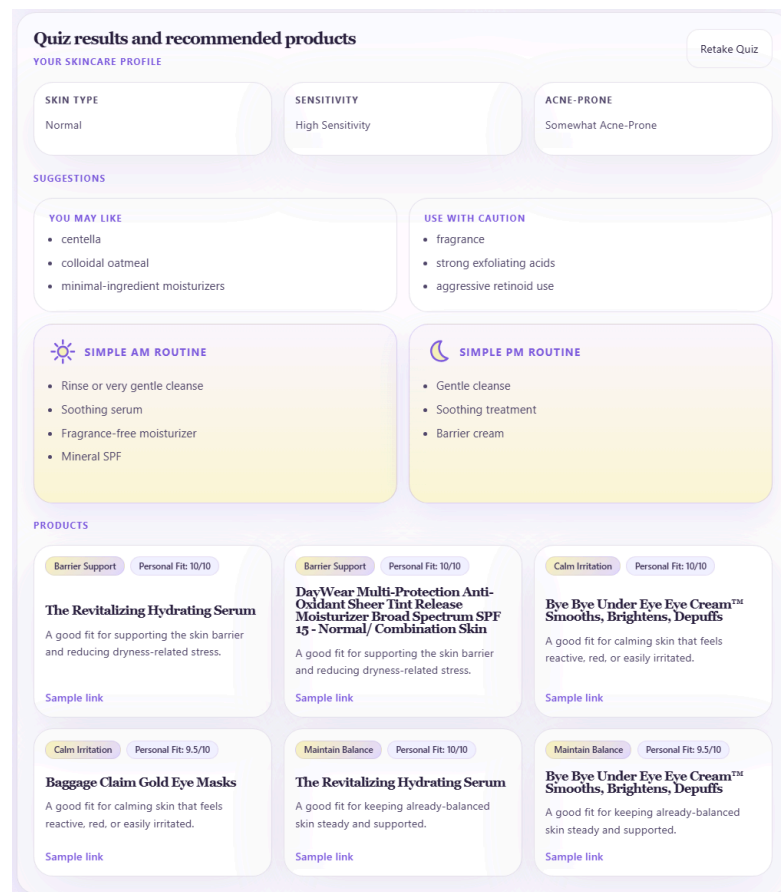


Figure 20. User results screen displaying generated skin profile and personalized product recommendations.

Product recommendations are retrieved from the Supabase database, enabling the app to fetch relevant products in real time. The system includes a fallback mechanism so that even if the database does not have perfect coverage for every user scenario, the app can still provide plausible suggestions.

Throughout development, care was taken to ensure the interface remained approachable and that any fallback messaging was clear and user-friendly.

Security and accessibility were also considered. Environment variables and credentials were handled securely, and basic accessibility needs were addressed—such as ensuring strong text contrast, readable colors, and clear navigation. The app was ultimately deployed using Vercel for public demonstration, and supporting documentation was created to make the project easier for others to understand and build on in the future.

Overall, the web app became a demonstration-ready tool that brought together the project's research, data, and recommendation logic in an accessible way for end users.

Considerations & Limitations

Several limitations affected how this project was conducted. One limitation was that Skin It to Win It had been developed with limited time and technology. As a result, the team achieved only a prototype during this period, rather than a complete software solution. Some features, such as usability testing, were omitted due to the limitations mentioned above.

Another limitation concerns the use of a survey as justification for this project. Although the survey data provided useful insight into skincare decision-making difficulties and the likely target audience, it cannot be considered fully representative, since respondents had specific preferences, experience levels, and skincare concerns that did not cover all possibilities.

A related limitation is associated with the recommendation algorithm used. In designing the prototype, the decision was made to use a rule-based system that could produce explainable recommendations rather than a medically informed diagnostic tool. While this decision helped in building an operational prototype, it also limited the scope of possibilities. As Chen argues, skincare recommendations are especially critical, as erroneous recommendations can lead to irritation, allergies, or deterioration of skin condition (456–57). Recommendations provided by this prototype, thus, can be seen simply as guidance rather than a medical diagnosis.

Lastly, the app has a valuable feature – social interaction. While the inclusion of this function adds to the app's value, it also brings some challenges. Sharing skincare experiences enriches the platform experience by fostering user engagement; however, peer recommendations are always subjective, so products that work for one person might be inappropriate for another. This function will need more polishing during future development.

Team Contributions

Lucy Lu

Lucy Lu was responsible for most of the project's design work. This included creating the low-fidelity sketch and refining it to a high-fidelity Figma prototype. She designed key screens for the application, including the home page, profile page, skin quiz results page, products page, product detail pages, and the community feature. In addition, she worked on the flow and layout of the skin quiz and the post layout in the community feature. Lucy also contributed substantially to the written report, particularly in sections related to design, literature review, methodology, and project framing.

Nicole Erin Keffer

Nicole Erin served as the web application developer for the project, leading the build and implementation of the final product. She developed the application end-to-end, including coding, styling, database integration, hosting, and testing, ensuring a fully functional and cohesive user experience.

Beyond development, she originated the capstone concept and played a central role in shaping the project's overall direction. She established structure across both group and individual work, helping maintain organization and clarity throughout the process, and supported team coordination through project management tools and workflow planning.

She also led the research and data components of the project, designing, distributing, and analyzing the project survey, and translating the findings into a supporting infographic. In addition, she sourced the datasets used in the project and contributed to initial organization and light data cleaning, ensuring the data was prepared for further processing and integration into the database.

Clio Torrance

Clio Torrance was the database engineer for the product, responsible for database design, database administration, and database population. Aside from the data itself, which was sourced from the International Nomenclature of Cosmetic Ingredients list and Sephora, the database is entirely her work. She refined her design as the project progressed, incorporating suggestions from other members, and handled backend data integration tasks. Clio also handled some progress presentations made to the instructor and other students during the design process.

Deliverables

Skin Quiz Question and Recommendation Logic

The skin quiz and recommendation logic were designed to enable this feature to serve as both an onboarding and personalization element for users. The quiz was designed around major factors affecting the skin, including oiliness, dryness, sensitivity, and acne. The results analysis logic was developed to help users understand their skin profile and define recommendation priorities.

Justification Survey + Infographic

A justification survey was designed and distributed to better understand user behaviors, knowledge gaps, and decision-making processes related to skincare. The survey collected data on factors such as product purchasing habits, level of research conducted before buying, confidence in selecting products, and understanding of ingredients.

The results were analyzed to identify key patterns and gaps in user knowledge, particularly around ingredient awareness and product selection confidence. These insights directly informed the direction and feature set of the Skin It to Win It platform, reinforcing the need for a more guided and educational skincare experience.

Findings were synthesized into an infographic that presents the research in a clear, accessible format. The infographic highlights key trends, user pain points, and opportunity areas, serving as both a supporting research artifact and a justification for the overall product concept.

Interactive High-Fidelity Prototype

An interactive high-fidelity prototype was created in Figma. The main objective of this prototype was to present to stakeholders and reviewers what the final product's UI/UX would look like. The prototype incorporated all key research and design decisions into a cohesive interactive model.

This prototype included the main screens of the Skin It to Win It app, including the homepage, skin quiz flow, skin quiz result screen, personal profile page, product search screen, product detail page, and community page. Those screens were not created as separate elements, but as integral parts of a user's journey within the app.

Skincare Product SQL Database

The skincare product database compiled real product data in a way that expressed the relationships between ingredients and products, which were necessary for recommendation logic and in-app informational displays. The database's structure was specifically designed for the demands of this project and required adaptation of existing data.

Web Application

The web application demonstrates the primary interactive experience for Skin It to Win It. Built with modern web technologies, the web app provides users with a clear, accessible way to receive product recommendations tailored to their skin needs. The application features a visually consistent interface, a step-by-step skin quiz for onboarding and personalization, and a results system that connects users to relevant product recommendations. Quiz responses are analyzed to generate a user skin profile, which then guides the recommendation logic behind the scenes.

In addition to real-time product recommendations, the web app includes fallback logic to ensure users always receive meaningful suggestions, even when database coverage is incomplete. The app's user interface was developed with accessibility and clarity in mind, and care was taken to ensure secure handling of environment variables and credentials. The final deliverable is a demonstration-ready, deployable web app that ties together the project's research, database, and recommendation logic, and is supported by documentation for future development or handoff.

Conclusions & Future Work

Skin It to Win It suggests that confusion about skincare could be addressed through information design, data management, personalization, and social interactions. The project, which included surveys, information design, database creation, and prototyping, presented a possible alternative to existing skincare apps.

A major contribution of the project lies in its approach to the issue: Skin It to Win It does not treat skincare management solely as a search task or as a recommendation process.

Nonetheless, the project still remains at an early stage. Possible areas of improvement include developing a more sophisticated algorithm for the recommendation system that moves beyond simple rule-based reasoning, increasing the volume and diversity of information about the products available on the site, and conducting more usability tests with a broader range of users. Moreover, while the social networking aspect of the project has been successful, improvements can be made, including clearer moderation guidelines, better post organization, and stronger linking of user profiles.

In conclusion, the project demonstrates strong potential for creating tools for skincare that are both data-driven and comprehensible, interactive, and socially aware. It has provided great insight into the development of consumer skincare resources.

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